Dynamic Survival Analysis on Consumer Loan Default:

Evidence from Prosper.com

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1. Motivation

-Survival analysis : model time to default
evaluate the credit risk and loan performance
(Banasik et al(1999), Bellotti and Crook(2007))

-Cox proportional hazard(1972) model is widely used

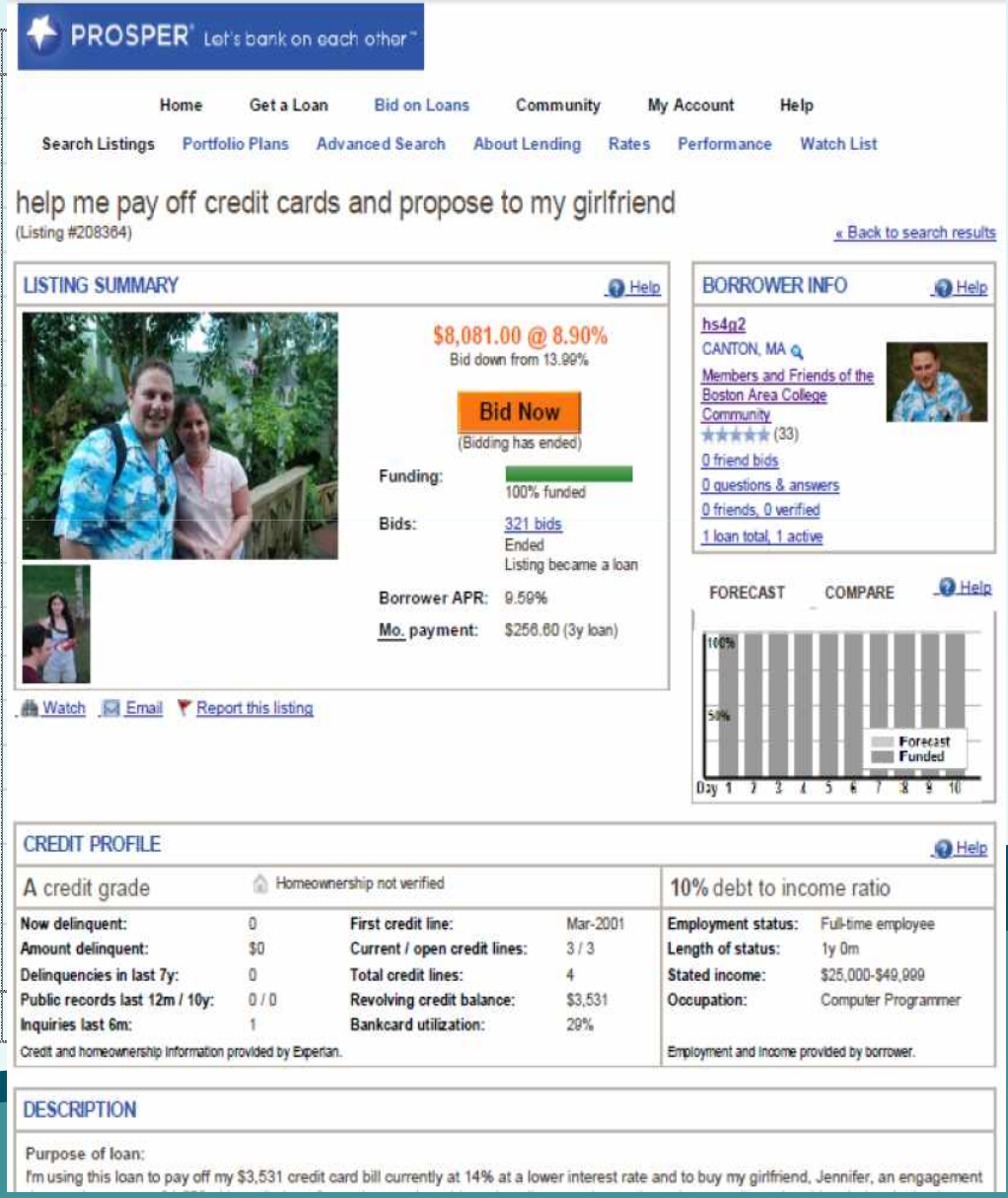
-But the assumption of Cox PH model is often violated
ex) constant effect vs time varying effect

-Time varying effect was not considered in the previous Consumer Loan Default study

- Estimate
(i) B-spline based Time varying coefficient Cox model
(Perperoglou et al 2006a/b)
(ii) Bayesian Time varying coefficient Cox model
(Wang et al, 2013).

2. Consumer loan default data

Variable Name	Description	Median
Amount	Dollar amount of loan	5000
AmountDelinquent	Dollar value of total delinquencies	0
BankcardUtilization	The percentage of available revolving credit that is utilized	0.61
BorrowerRate	Interest rate paid by borrower	16.30
CityDummy	1 if the borrower is a resident of a city	0
CreditScore	Credit Score of a borrower evaluated by credit bureau	670
CurrentCreditLines	Number of accounts that the borrower is paying on time	8
CurrentDelinquencies	Number of credit lines currently listed as delinquent	0
DebtToIncomeRatio	Outstanding debt / self-reported income	0.21
DelinquenciesLast7Years	Number of delinquent accounts in the last seven years	0
Fulltime	1 if the borrower reports they are a full-time employee	1
GroupDummy	1 if the borrower is a member of a Prosper social group	0
Home	1 if the borrower is a homeowner	0
InquiriesLast6Months	Number of times the credit report has been	1
OpenCreditLines	Number of active lines of credit in the borrower's name	7
PublicRecordsLast10Years	Number of public records such as bankruptcies or liens	0
PublicRecordsLast12Months	Number of public records	0
RevolvingCreditBalance	Outstanding balance on credit cards or other revolving credit	5122
TotalCreditLines	Total number of credit lines listed in the borrower's credit report	22
Sample size = 17,903		



3. Cox PH model

- $h(t|X_i) = h_0(t)\, exp(X_i'\beta) = h_0(t)\, exp(\beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})$ for all i=1,2,…n

	coef	exp(coef)	se(coef)	z	Pr(>|z|)	
CreditScore	-0.003	0.997	0.003	-1.079	0.281	
CreditScore2	0.000	1.000	0.000	0.272	0.786	
GroupDummy	0.045	1.046	0.030	1.496	0.135	
Amount	0.000	1.000	0.000	16.485	< 2e-16	***
BorrowerRate	0.181	1.199	0.013	14.123	< 2e-16	***
BorrowerRate2	-0.003	0.997	0.000	-10.912	< 2e-16	***
DebtToIncomeRatio	0.040	1.041	0.011	3.678	0.000	***
Home	0.210	1.234	0.030	7.042	0.000	***
CurrentDelinquencies	0.040	1.041	0.005	8.173	0.000	***
DelinquenciesLast7Years	-0.005	0.995	0.001	-4.316	0.000	***
InquiriesLast6Months	0.052	1.053	0.003	19.565	< 2e-16	***
PublicRecordsLast10Years	0.033	1.033	0.013	2.521	0.012	**
PublicRecordsLast12Months	-0.078	0.925	0.050	-1.551	0.121	
OpenCreditLines	-0.019	0.981	0.008	-2.337	0.019	**
CityDummy	-0.144	0.866	0.039	-3.734	0.000	***
Fulltime	-0.107	0.899	0.040	-2.691	0.007	***
BankcardUtilization	-0.176	0.839	0.040	-4.442	0.000	***
RevolvingCreditBalance	0.000	1.000	0.000	0.539	0.590	
AmountDelinquent	0.000	1.000	0.000	-0.724	0.469	
TotalCreditLines	0.001	1.001	0.001	0.607	0.544	
CurrentCreditLines	0.011	1.011	0.008	1.467	0.142	

4. Does PH assumption hold?

- Scaled Schoenfeld residual test: *correlation* between the *covariate-specific residual* and *event times*

Proportional Hazard Test (H_0: constant coefficient)

	rho	chisq	p	
CreditScore	0.011	0.767	0.381	
CreditScore2	-0.011	0.774	0.379	
GroupDummy	0.027	4.390	0.036	**
Amount	0.001	0.003	0.954	
BorrowerRate	-0.037	9.100	0.003	***
BorrowerRate2	0.020	2.650	0.103	
DebtToIncomeRatio	0.012	0.920	0.337	
Home	0.043	11.000	0.001	***
CurrentDelinquencies	-0.033	6.540	0.011	**
DelinquenciesLast7Years	0.026	4.630	0.031	**
InquiriesLast6Months	-0.001	0.002	0.963	
PublicRecordsLast10Years	0.031	5.610	0.018	**
PublicRecordsLast12Months	0.006	0.242	0.623	
OpenCreditLines	-0.027	4.940	0.026	**
CityDummy	0.024	3.680	0.055	*
Fulltime	0.001	0.012	0.912	
BankcardUtilization	0.039	10.100	0.001	***
RevolvingCreditBalance	-0.019	2.320	0.127	
AmountDelinquent	-0.010	1.000	0.317	
TotalCreditLines	0.011	0.831	0.362	
CurrentCreditLines	0.035	8.070	0.004	**
GLOBAL	NA	207.000	0.000	***

5. B-spline based time varying coef. Cox model

$$h(t \mid X_i) = h_0(t)\, exp(X_i' \beta(t))$$

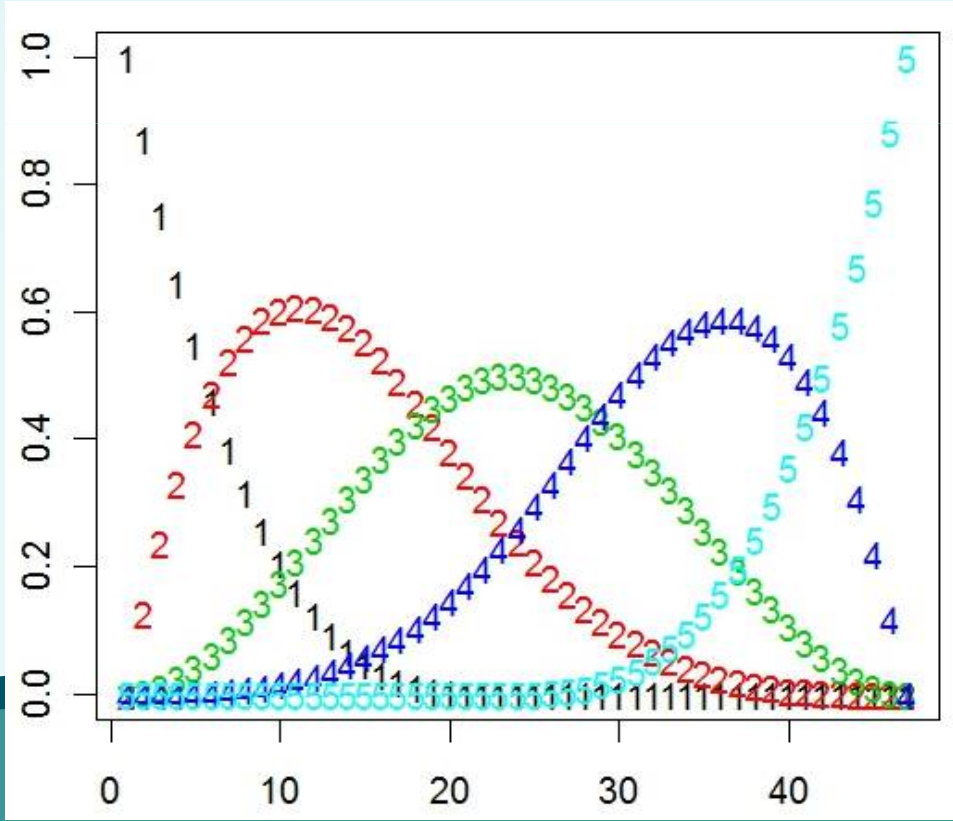
$$h(t \mid X_i) = h_0(t)\, exp(X_i\, \Theta\, F'(t_i))$$

where $\Theta_{p \times q} = (\beta_1\ \beta_2 \ldots \beta_q)$ is a matrix of coefficients and $F(t) = (f_1(t)\ f_2(t) \ldots f_q(t))$ is time function.

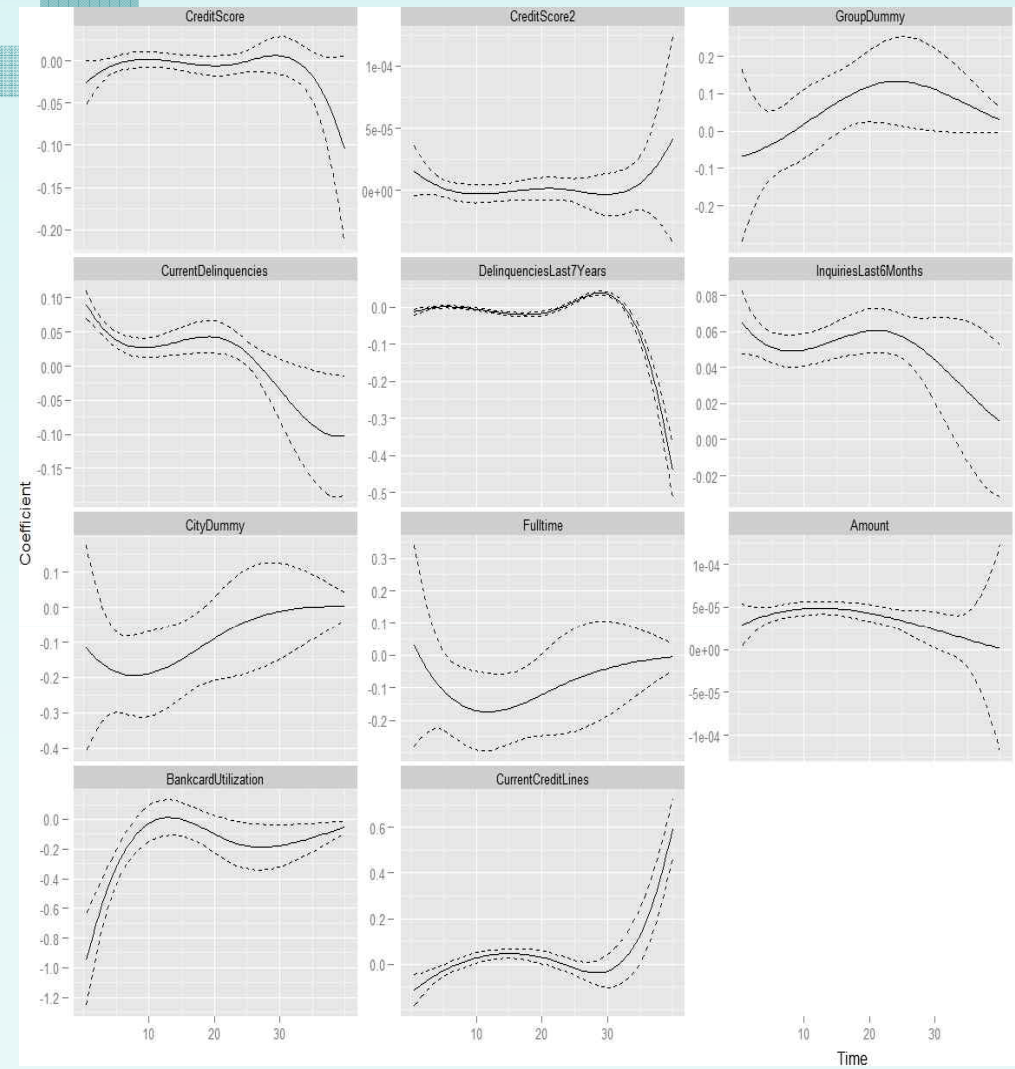
matrix Θ can be written as a product of two sub-matrices, $B_{p \times r}$ and $\Gamma'_{r \times q}$,

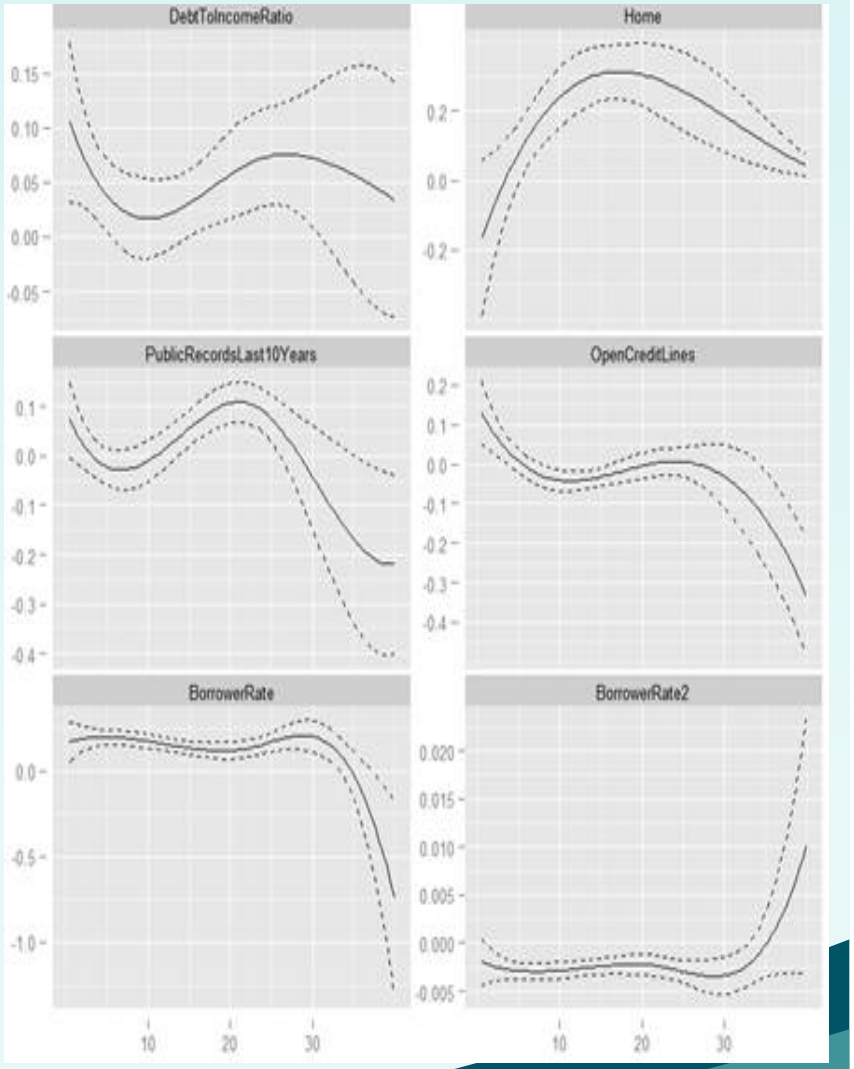
Akaike's information criterion(AIC) to define the optimum rank

F(t) : B-spline with d.f.=5
(cubic spline with 1 knots
(t=23) and constant)

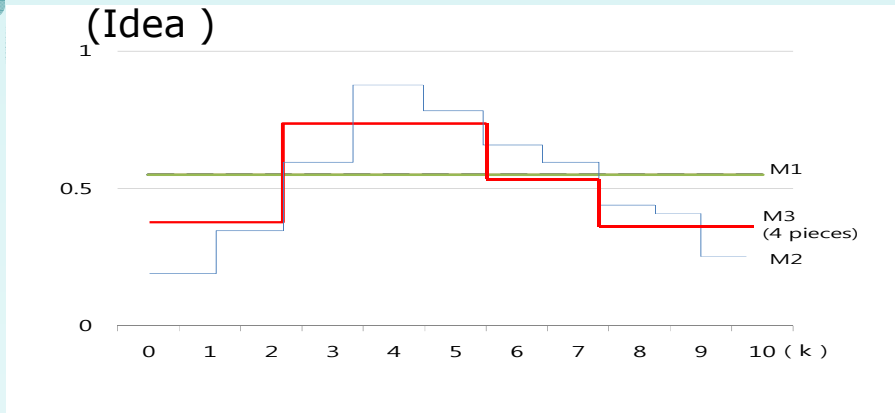


(Estimation result)



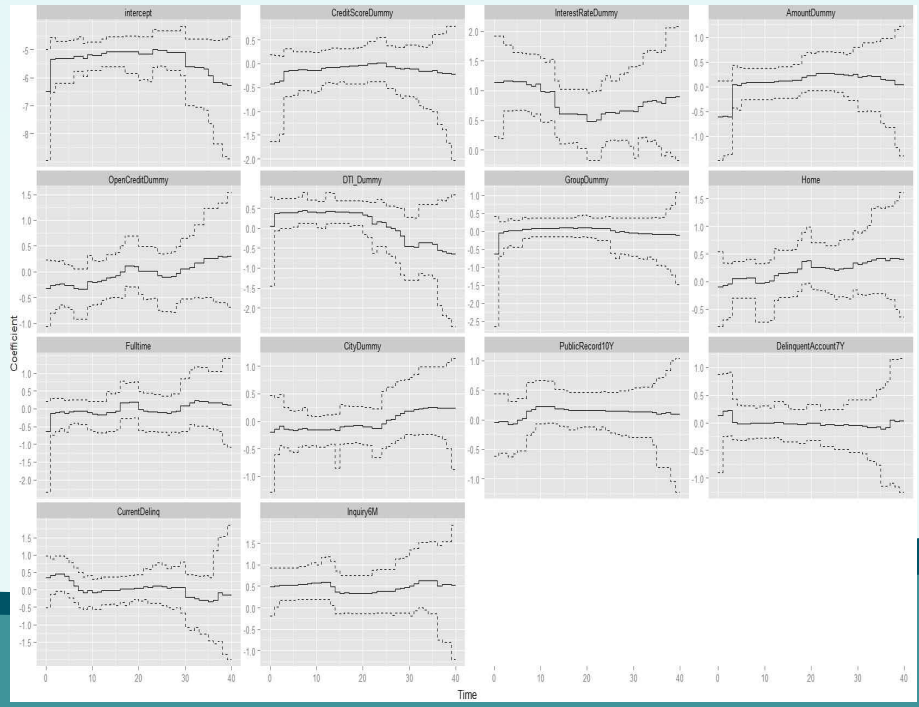
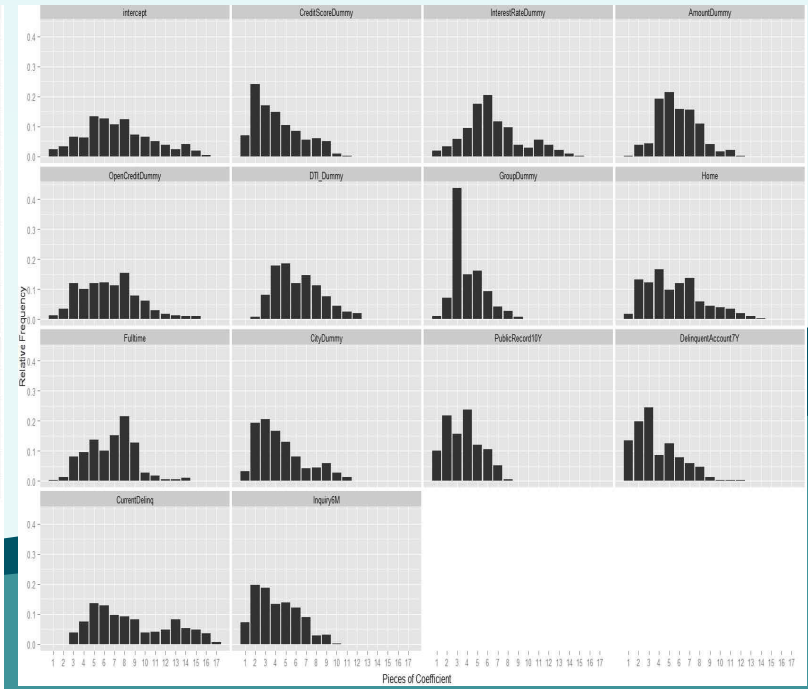


6. Bayesian time varying coef. Cox model



- RJ MCMC method is used
- only admit dummy covariates
- used simplified data

▪ Bayesian time varying coef. Cox model (iteration=3,000, burn in=1,000, thin=1)



▪ Histogram of the number of Jump (iteration =3,000, burn in=1,000, thin=1)